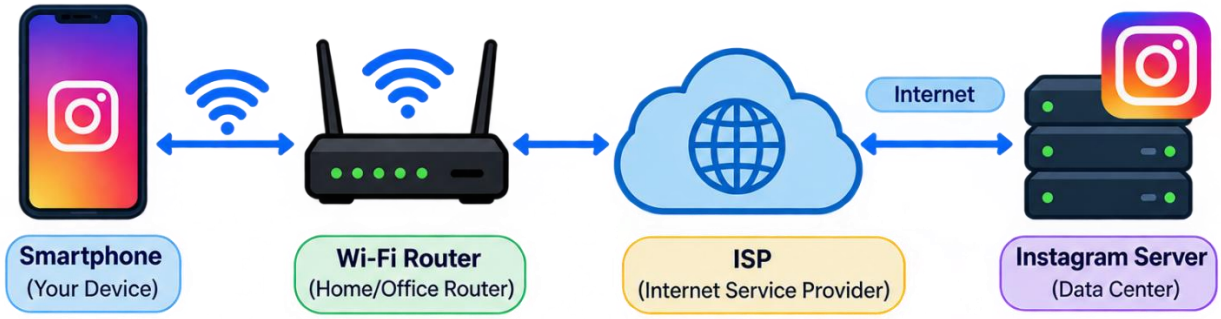
Networking technologies

Tasks

1. Draw a simple network diagram showing how your smartphone connects to Instagram — include your device, Wi-Fi router, ISP, and Instagram's server, and label each part.

Ans.:-



1 Smartphone
Your smartphone connects to the Wi-Fi network and sends a request to Instagram's server.

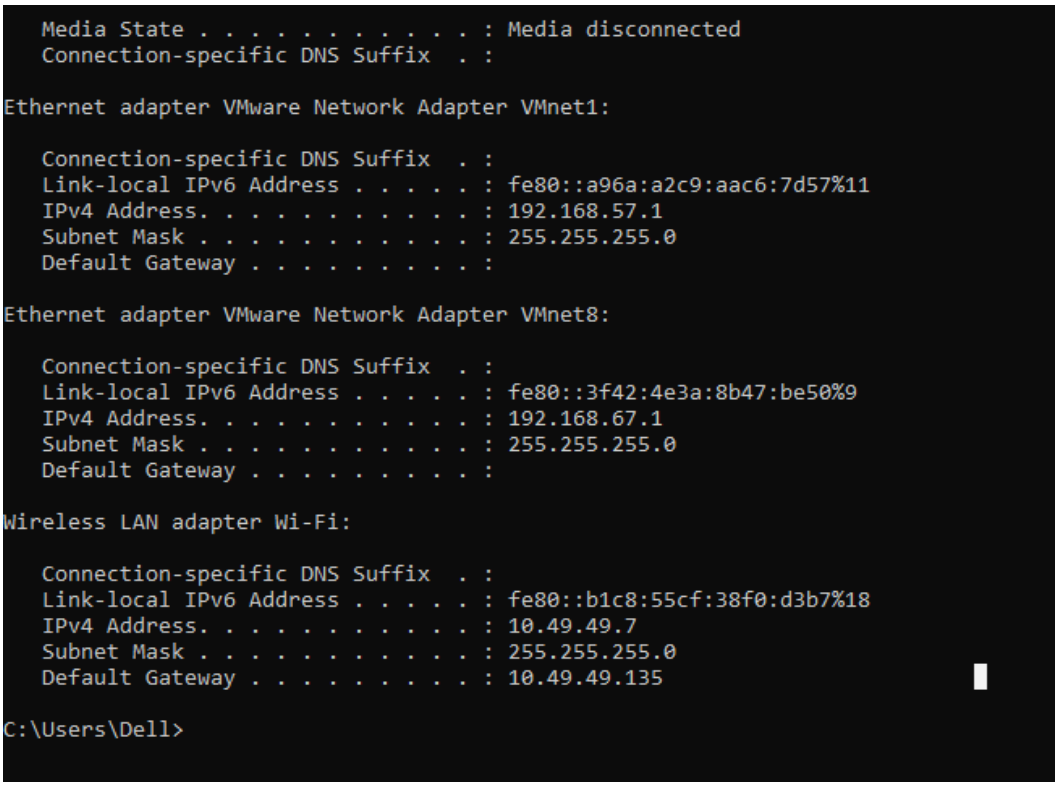
2 Wi-Fi Router
The router connects your device to the internet and forwards your data to your ISP.

3 ISP
Your Internet Service Provider routes your request through the internet to Instagram's network.

4 Instagram Server
The request reaches Instagram's data center, which sends back the required data (photos, videos, messages, etc.) to your device.

2. List the IP address of your device and your default gateway using the 'ipconfig' command (Windows) or 'ifconfig' (Linux/Mac), and explain what each means in 1-2 lines.

Ans.:-



Item	Example	Meaning
IP Address	192.168.67.1	The unique address assigned to your device on the local network.
Default Gateway	192.168.66.1	Usually your Wi-Fi router's IP address; it provides the path from your local network to other networks/Internet.

**3. Simulate a Zomato food order journey by writing the steps/data packets that travel from your device to Zomato's server and back, mentioning protocols like HTTP, TCP/IP, and DNS wherever relevant.

Hint: Think about what happens when you search for a restaurant and place an order.**

Ans.:-

Zomato Food Order Journey – Data Packets

Imagine you search for a restaurant on the Zomato app and place an order:

1. Open Zomato

- Your smartphone connects to Wi-Fi/mobile network.
- The device gets an IP address and uses TCP/IP to communicate over the Internet.

2. DNS Request

- Your device needs the IP address of Zomato's server.
- It sends a **DNS query** for the Zomato domain.
- DNS returns the server's IP address.

3. Search for a Restaurant

- Your app sends a request to Zomato's server using HTTPS (HTTP over TLS).
- TCP establishes a reliable connection and IP handles addressing/routing.
- The request contains information such as your search term and location-related data needed for the search.

4. Zomato Server Responds

- Zomato's server processes the request.
- Restaurant information is sent back through the Internet using TCP/IP + HTTPS.
- Your app displays the restaurants.

5. Place the Order

- You select food, enter/confirm delivery details, and click Place Order.
- The app sends the order information to Zomato's server using HTTPS.

6. Server Processes the Order

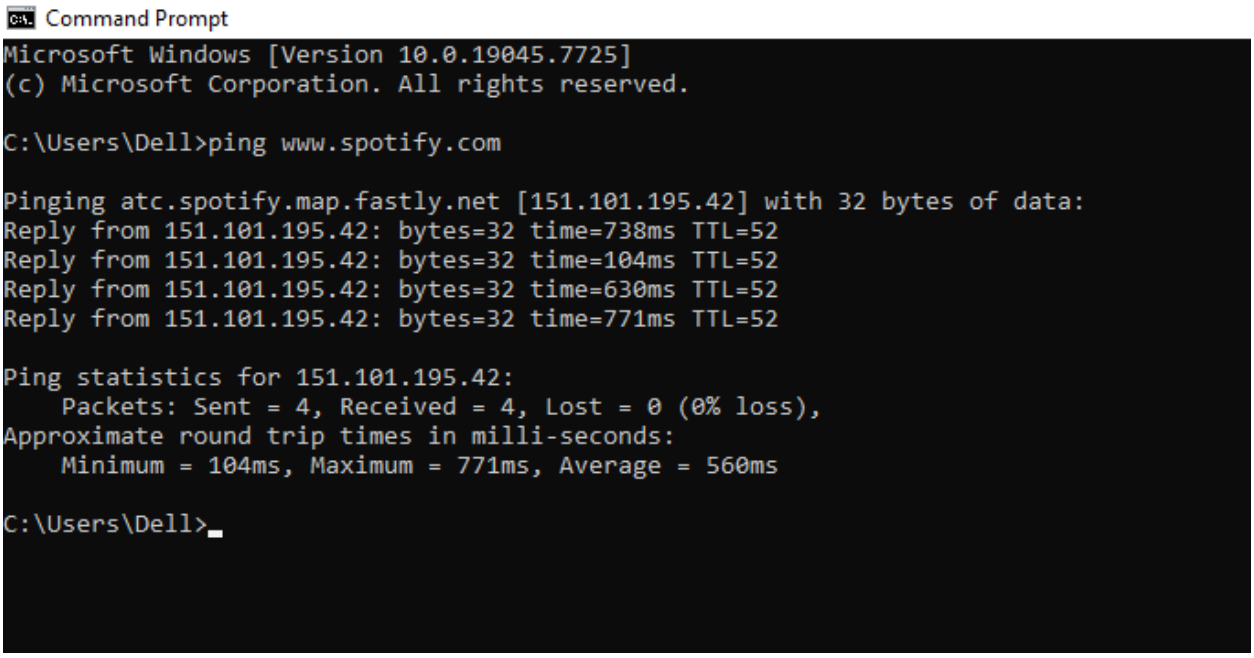
- Zomato's server sends the order to the restaurant and processes payment/order status.
- Responses travel back to your device through TCP/IP.

7. Order Confirmation

- Your phone receives the confirmation.
- The app displays something like “Order Confirmed” and later shows delivery updates.

4. Use the 'ping' command to check the connection between your computer and www.spotify.com, then explain what the results mean for network troubleshooting.

Ans.:-



The ping command checks whether a destination is reachable and measures network response time and packet loss. It is useful for identifying connectivity problems, high latency, or packet loss during network troubleshooting.